



Generation Interconnection

Queue Project AD2-166

Ironwood 230 kV

System Impact Study Report

Capacity : 89.8 MW / Energy : 75 MW

June 2019

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Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

PJM has timely issued this System Impact Study without ICTR determinations. PJM will work with each customer to identify the Customer-Funded Upgrade(s) and LDAs (no more than three) for which the customer wants PJM to determine ICTRs. PJM will provide that determination as quickly as practicable following issuance of this System Impact Study.

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, Section 205, as well as the System Impact Study Agreement between Helix Ironwood, LLC, the Interconnection Customer (IC) and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Mid-Atlantic Interstate Transmission (MAIT or METED).

General

The IC has proposed an uprate to its existing natural gas-fired 2 x 1 generation facility, designated PJM Queue Numbers A01 and AC1-035, located at 305 Prescott Road in Lebanon, Lebanon County, Pennsylvania. The increased capability of AD2-166 Queue Request is achieved through the installation of additional chillers and upgrading old equipment. The total capability of this project will be 825 MW. PJM recognizes 780 MW as Capacity Interconnection Rights.

The following table shows the MW net energy contributions to the Maximum Facility Output (MFO) and the total Capacity Interconnection Rights (CIR) by Queue No.:

	Queue No.	Net Energy MW Contribution	Capacity Interconnection Rights
	A01	720	660.2 [1]
	AC1-035	30	30
	AD2-166	75	89.8
Total MW		825 (MFO) [2]	780 (CIR) [3]

[1] Above 660.2 MW CIR is based on the generators summer capability tests effective 9/1/2017@0000 hours

[2] MFO stands for Maximum Facility Output which is a defined Tariff term.

[3] CIR stands for Capacity Interconnection Rights which is a defined Tariff term.

The expected Commercial Operation Date is **April 30, 2022**.

The following is a listing of all study reports and agreements by queue # and where to find them at PJM website:

Queue No.	Type of Document	PJM Website Link
A01	System Impact Study	https://www.pjm.com/pub/planning/project-queues/impact_studies/a01_imp.pdf
A01	Facilities Study	https://www.pjm.com/pub/planning/project-queues/facilities/a01_fac.pdf
AC1-035	System Impact Study	https://www.pjm.com/pub/planning/project-queues/impact_studies/ac1035_imp.pdf

AC1-035	Interconnection Service Agreement (ISA)	https://www.pjm.com/pub/planning/project-queues/isa/ac1_035_isa.pdf
AC1-035	Interconnection Construction Service Agreement (CSA)	https://www.pjm.com/pub/planning/project-queues/csa/ac1_035_csa.pdf

Point of Interconnection

The Point of Interconnection remains unchanged as in A01 and AC1-035 queue projects. The Helix Ironwood facility interconnects with the MAIT transmission system via split POI as shown in the one-line diagram in Appendix 2.

The facility is injecting power directly into the Ironwood substation which connects to the South Lebanon – North Lebanon 230 kV transmission line.

Cost Summary

This project is an increase to an existing direct connection. Therefore, there are no costs for direct and non-direct connection facilities. The AD2-166 project does not have any system reinforcements.

Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FirstEnergy’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of fully rated 230 kV circuit breaker to protect the AD2-166 generator lead line. A single breaker must be used to protect this line. FirstEnergy does not approve for individual GSU transformer breakers (if applicable) to protect this line.
2. The purchase and installation of the minimum required FirstEnergy generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (SCADA) equipment to provide information in a compatible format to the FirstEnergy Transmission System Control Center.
4. The execution of a back-up service agreement to serve the customer load supplied from the AD2-166 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.
5. Power factor and voltage control requirements: The IC is required to comply with the FE and PJM generator power factor and voltage control requirements. The existing 720 MW (Queue #A01) portion of the Customer Facility shall retain its existing ability to maintain a power factor of at least

0.95 leading (absorbing VARs) to 0.90 lagging (supplying VARs) measured at the generator's terminals.

The increase of 30 MW (Queue #AC1-035) shall be designed with the ability to maintain a power factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals. The increase of 75 MW (Queue #AD2-166) shall be designed with the ability to maintain a power factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals.

Based on unit data provided by the IC, the Customer Facility is capable of meeting the power factor requirements.

6. System Protection Requirements: The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FirstEnergy's "Requirements for Transmission Connected Facilities" document located online at above shown link. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.
7. The IC must meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures. For example, the IC will need to properly locate and report the over and under-voltage and over and under-frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FirstEnergy system.

The above FirstEnergy requirements apply in addition to any metering or other requirements of PJM.

Revenue Metering and SCADA Requirements

PJM Requirements

The IC will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

First Energy Requirements

The IC will be required to comply with all FirstEnergy revenue metering requirements for generation interconnection customers which can be found in FirstEnergy's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

Network Impacts

The Queue Project AD2-166 was evaluated as a 75.0 MW (Capacity 89.1 MW) injection at the Ironwood 230 kV substation in the Met-Ed service area. Project AD2-166 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-166 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

PJM performed a power flow analysis of the transmission system using a 2021 summer peak load flow model and the results were verified by FirstEnergy. Additionally, FirstEnergy performed an analysis of its underlying transmission <100 kV system. The AD2-166 project did not contribute to any overloads on the FirstEnergy transmission system

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Short Circuit

None. (No overdutied circuit breakers identified)

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection

Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

System Reinforcements

Short Circuit

None.

Stability and Reactive Power Requirement

A dynamic stability analysis was completed by PJM and the results were reviewed by FirstEnergy.

The AD2-166 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 15-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2021 Light Load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.

- b) The AD2-166 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

AD2-166 meets the power factor requirements.

Reactive Power Assessment

The reactive power capability of AD2-166 meets the power factor requirement at the generator terminal as shown in the following table:

Generator	MFO (MW)	Required Power Factor Range		Maximum Lagging (MVar)	Minimum Leading (MVar)
		Lagging	Leading		
Existing	720	0.9	0.95	348	-237
AC1-035	30	0.9	1	15	0
AD2-166	75	0.9	1	36	0
Total Reactive Power Required				399	-237
Reactive Power from Generator				Qmax	Qmin
				474	-380
CT1				155	-125
CT2				155	-125
ST				164	-130
Customer Planned Compensation				0	0
Deficiency in Reactive Power				Meet	Meet

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

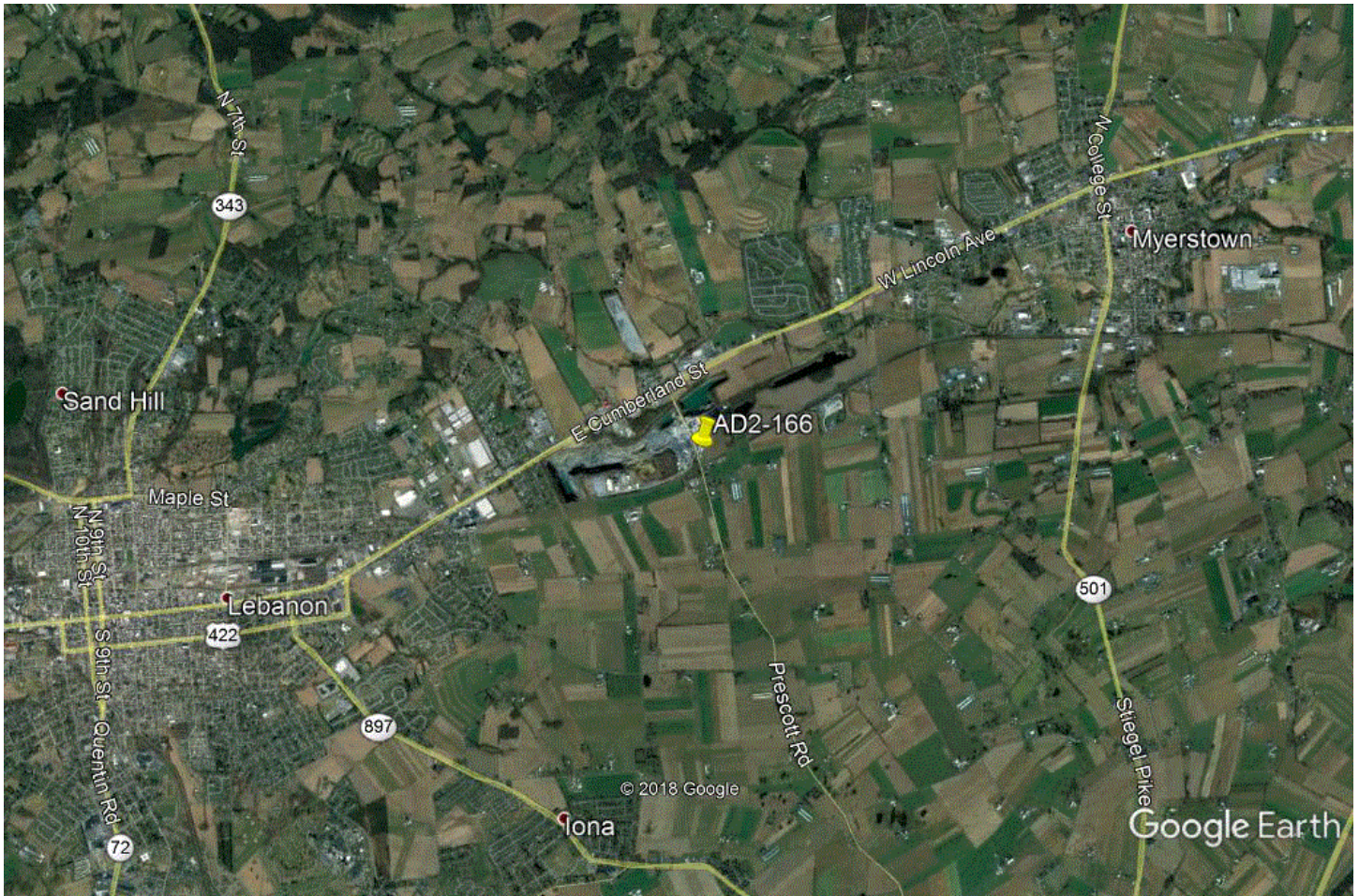
Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None.

Appendix 1: Facility Location

PJM Queue Position: AD2-166



Appendix 2: One-Line Diagram

PJM Queue Position: AD2-166

